Feature Article

Polymer characterization by temperature gradient interaction chromatography

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SUMMARY: Thermodynamic principles and some applications of the temperature gradient interaction chromatography (TGIC) recently developed for the characterization of synthetic polymers are described. TGIC is a form of high performance liquid chromatography (HPLC) that varies column temperature in a programmed manner to control the retention of polymeric species during isocratic elution. The retention of polymers strongly depends on their molecular weights, and the polymers are well separated by TGIC in terms of their molecular weights. TGIC is superior to size exclusion chromatography (SEC) in resolution and sample loading capacity, and has higher sensitivity to molecular weight in the analysis of nonlinear polymers. TGIC has an advantage over solvent gradient HPLC because it permits the use of refractive index sensitive detection method such as differential refractometry and light scattering due to the isocratic elution. In addition, temperature provides finer and more reproducible retention control than the solvent composition, which is important in determining the molecular weight distribution by secondary calibration methods. With TGIC analysis, we found that the molecular weight distribution of anionically polymerized polymers is much narrower than has been generally accepted from SEC analysis. We also found the TGIC separation conditions for polystyrene, polyisoprene, poly(methyl methacrylate), poly(vinyl chloride), and poly(vinyl acetate) over wide molecular weight range. Because of its sensitivity to the molecular weight alone, TGIC was successfully applied to the characterization of star shaped polystyrene, and the detailed linking kinetics between living polystyrene anions and a chlorosilane linking agent was investigated.

Introduction

Liquid chromatography (LC) is a powerful tool for the characterization of natural and synthetic macromolecules which are often heterogeneous in molecular weight and/or in chemical composition. Among numerous variations of LC methods, size exclusion chromatography (SEC) has been the most popular method for the analysis of the molecular weight distribution of synthetic polymers[1–3]. It allows us to characterize the molecular weight distribution of polydisperse polymers with relative ease compared to the other classical fractionation techniques with respect to time, effort, and required amount of sample. SEC fractionates polymer molecules by the partition equilibrium of polymer chains between a common solvent but located at the interstitial space and the pores of the column packing materials, typically in the form of uniform size porous beads. Therefore the partition equilibrium is mainly governed by the conformational entropy difference of the polymer chains in two different physical environments. The eluent is chosen to minimize the enthalpic interaction between the polymeric solutes and the stationary phase. In results, SEC separates the polymer molecules in terms of the volume occupied by a single polymer chain in the elution solvent. If there exists a simple relationship between the polymer chain volume and its molecular weight such as for linear and chemically homogeneous polymer molecules, retention volume of a polymer molecule is well correlated with its molecular weight. However, the same cannot be said for nonlinear chain polymers and copolymers in which the relationship between the molecular weight and the molecular size is more complicated. SEC is not able to separate them according to their molecular weight. Rigorous characterization of branching in synthetic polymers still remains as one of the major challenges in the SEC analysis of polymers[4].

Since SEC separates the polymer molecules according to the size only, it is not an efficient tool to separate the molecules in terms of chemical heterogeneity, such as chemical composition differences of copolymers, tacticity, and end-group difference. Interaction chromatography(IC) is suitable for the purpose since the technique is sensitive to the chemical nature of the molecules. In contrast to SEC, IC utilizes mainly the enthalpic interaction, adsorption or partition of solute molecules to the stationary phase. As well documented in the literature, IC is an

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established tool to separate polymers efficiently in terms of chemical composition, tacticity, end groups, and so on[5–7].

In the last two decades, there also have been a number of reports on the use of IC for the fractionation of polymers in terms of molecular weight. In the early years, these separations were done by thin-layer chromatography with a silica gel stationary phase[8–11]. Many of these separations suffered from relatively poor resolution, particularly for high molecular weight polymers. The separation efficiency was greatly improved by employing the reversed phase stationary phases. Armstrong and Bui were the first to separate polystyrenes successfully in terms of molecular weight by reversed phase thin layer as well as column chromatography[12]. Subsequently a number of reports on the IC fractionation of high polymers, polystyrene[13–16], poly(methyl methacrylate)[17], poly(ethylene oxide)[18], have followed.

All of these HPLC fractionations employed the solvent gradient elution method. Since the retention of the polymer molecules strongly depends on molecular weight, it is necessary to control the retention during the elution in order to deal with the wide molecular weight distribution of synthetic polymers. Boehm et al. proposed a model called "critical solution theory" for the retention behavior found in the isocratic elution of polymers. According to the theory, the fractionation of any molecular weight range of polymers can be only accomplished by altering the solvent composition in the mobile phase. Therefore, isocratic elution should be either impractical or impossible for wide ranges of molecular weight[19, 20]. On the other hand, Snyder et al. assert that no special model is needed for the polymer retention and the traditional models can be used for the polymers as well[21]. Lochmüller and McGranaghan demonstrated that the isocratic elution of high molecular weight PS is possible when they used a precolumn mixer[13]. Alhedai et al. also have reported the isocratic elution behavior of polystyrene[14], and the issue of isocratic IC elution of polymers seems to be no longer under debate although further studies need to be done to elucidate a general separation mechanism of high polymers[22]. According to our experience, as described later in more detail, retention behavior of polymers is not much different from small molecules if one considers the inherent molecular weight distribution of synthetic polymers and the large number of repeating units in a single polymer molecule. In addition, conventional injection systems without any special pre-mixer works well if we use the same injection solvent as the eluent.

The retention in IC is generally expressed with the capacity factor (k') which is defined as follows:

$$k' \equiv K\frac{V_s}{V_m} = \frac{t_R - t_m}{t_m} \tag{1}$$

where K is the partition coefficient of a solute between the stationary and mobile phase, t_R is the retention time of the solute, t_m is the time for the eluent to sweep the total mobile phase volume (commonly determined from the retention time of the injection solvent), and V_s and V_m are the volumes of the stationary and mobile phase, respectively. K is related to the standard Gibbs free energy change (ΔG^c) of the sorption process, and ΔG^c can be further divided into the enthalpic and entropic contribution of the sorption process.

$$K = \frac{C_s}{C_m} = \exp\left(\frac{\Delta G^0}{RT}\right) = \exp\left(\frac{\Delta S^0}{R} - \frac{\Delta H^0}{RT}\right) \tag{2}$$

The exponential relationship between k' and ΔG^0 indicates that the retention of solutes varies much if ΔG^0 of the solutes of interest spans a wide range, which in turn makes an isocratic elution difficult. The same problem exists in the IC analysis of synthetic homopolymers. According to the well known empirical observation, the so-called Martin's rule, ΔG^0 is proportional to the degree of polymerization[5]. Therefore K (also k') changes exponentially with the degree of polymerization of the polymer molecules. In order for a polymer sample with a wide molecular weight distribution to be eluted in a reasonable experimental time period, it is necessary to change K during the elution. One way to control the retention is to change ΔG^0 itself. Solvent gradient elution is a method to change ΔG^0, thus k', by changing the eluent composition during the elution in a pre-programmed manner. The solvent gradient HPLC fractionation of synthetic polymers works well, but it has a critical disadvantage in the analysis of the molecular weight distribution of a polymer, since it is difficult to use many useful detection methods for the characterization of polymers, such as differential refractometry, light scattering, and viscometry due to the background signal drift. Unlike the analysis of small molecules which are unique chemical species in general, synthetic polymers are collections of homologue series of molecules of different degree of polymerization and their molecular weight distribution is one of the most important parameters to determine their ultimate properties. In order to characterize the molecular weight distribution of a synthetic polymer, it is necessary to quantitatively determine the relative abundance of the species with different degree of polymerization. The background signal drift makes the quantitative analysis difficult, if not impossible.

An alternative method to control the retention is to change the temperature, as easily inferred from Eq. (2). It is well known that temperature can alter the retention behavior of IC. In fact, temperature dependency of k' has been widely used to study the thermodynamics involved in the HPLC separation process. However, temperature has not been a popular variable in practical IC separations, mainly because the useful range of temperature variation is limited by the freezing point of the mobile phase (or precipitation of the polymeric solutes) at the low end and the boiling point at the upper end. Therefore temperature as a control variable is not as effective as solvent composition if the sample contains components over a large ΔG^0 range such as in the analysis of a sample containing many different chemical species. In the IC analysis of synthetic polymers, temperature has been suggested but not practiced as a control variable of the retention until Lochmüller et al. first reported a systematic study on the temperature effect of the reversed-phase HPLC separation of poly(ethylene glycol)[18]. About the same time, we also found independently that temperature provides a very efficient retention control in reversed-phase HPLC to fractionate polystyrene[23]. In this feature article, we will briefly review our recent effort on temperature gradient interaction chromatography (TGIC) separation of synthetic polymers.

Experimental part

The chromatographic apparatus for TGIC is a typical reversed-phase HPLC instrument except for a modification to precisely control the temperature of the separation column. The schematic diagram of the apparatus is shown in Fig. 1. Temperature control of the separation column is done by circulating a fluid from a bath/circulator (Neslab, RTE-111) through a column jacket. In order to minimize the temperature gradient across the diameter of the column, the temperature of the eluent is pre-equilibrated with the column temperature by passing through an empty column (or a long piece of tubing) of which the temperature is controlled by another column jacket. All experiments were carried out with a typical HPLC system equipped with C18 bonded silica column(s) with various pore sizes. In addition to commercial columns, home packed columns were often used. These columns were packed in empty stainless columns for 40 min at 7500 psi with a slurry packer (Alltech, Model 1666). The slurry was made by mixing 4 g of packing material with 20mL of methanol, then degassed by sonication for 1 min.

When a mixed eluent was used, HPLC grade solvents were premixed to a desired composition and delivered in isocratic mode. We found that the pre-mixing gave more reproducible results than mixing the eluent by use of a gradient pump. Samples for injection were dissolved in a small portion of the eluent in the chromatographic run unless the sample did not dissolve in the eluent at the room temperature. The typical concentration of injected samples was 1 mg/mL except for light scattering detection. In order to obtain adequate signal to noise ratio of low molecular weight samples for light scattering detection, the sample concentration was adjusted appropriately depending on the molecular weight. Sample solutions were filtered by 0.45 μm PTFE membrane disk filters (Gelman) and then injected through 6-port Rheodyne 7125 injector. Depending on the polymers and eluent system, an effective combination of a UV/Vis detector (LDC Analytical, Spectromonitor 3200), a differential refractometer (RI) detector (Waters, R401 or Shodex RI71), low angle laser light scattering (LALLS) detector (LDC Analytical, KMX-6), and an evaporative light scattering (ELS) detector (Polymer Lab., PL-EMD950) were used to record the chromatograms.

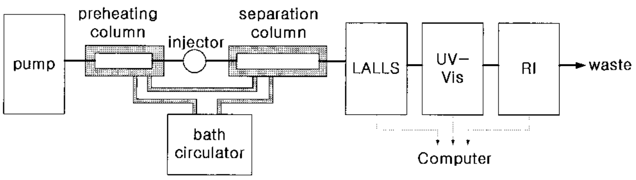

Fig. 1. Schematic diagram of the TGIC apparatus

Results and discussion

Solvent and temperature effect on IC separation of synthetic polymers

Fig. 2 shows an example of the solvent effect in the IC elution of polystyrenes (PS) where 7 chromatograms of a mixture of 6 polystyrene (PS) standards are displayed. All the chromatograms were obtained at the constant temperature of 30.5 °C, but with the mixed eluents of CH_2Cl_2/CH_3CN at different compositions, where CH_2Cl_2 is a good solvent and CH_3CN is a poor solvent for PS. Each chromatogram is labeled with the corresponding vol.-% of CH_2Cl_2 in the eluent. At high CH_3CN composition, the interaction of PS with the C18 bonded silica stationary phase is strong and the polymers are retained strongly in the column. With the increase of the molecular weight of PS, the retention increases rapidly as expected from Eqs. (1) and (2), and the elution peak broadens significantly. As the fraction of CH_2Cl_2 increases, retention of PS becomes shorter. This effect is used in the solvent gradient elution method to control the retention of the solutes during the IC elution. At the composition of 57/43 (v/v) CH_2Cl_2/CH_3CN, all the PS standards are eluted as a single peak. This is the *chromatographic critical point* or the *adsorption-size exclusion transition point* where the enthalpic and entropic terms in Eq. (2) compensate each other[24,25]. This compensation results in the loss of separation selectivity according to the polymer molecular weight, and all the PS standards are eluted near the injection solvent peak position regardless of their molecular weights. At higher CH_2Cl_2 composition, all the peaks shift to shorter t_R and the PS standards are eluted before the injection solvent peak. In this separation condition, PS samples are separated mainly by the size exclusion mechanism. We can also note that the elution sequence is reversed and the higher molecular

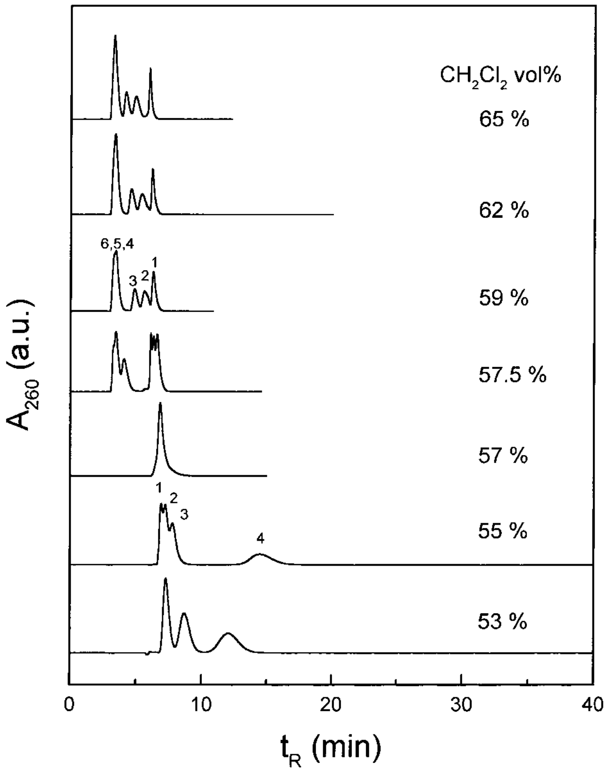

Fig. 2. Solvent composition effect on the retention of 6 PS standards ($\overline{M}_w$: (1) 2.5 k, (2) 12 k, (3) 29 k, (4) 165 k, (5) 502 k, (6) 1800 k (k means factor 10^3). The chromatograms were obtained at the column temperature of 30.5 °C with the mixed eluents of CH_2Cl_2/CH_3CN at different compositions as labeled in the figure. Injection sample concentration was 1 mg/mL of each PS standards and the injection volume was 50 μL. Single C18 bonded column (Alltech, Nucleosil, 250 × 4.6 mm, 100 Å pore size, 5 μm particle size) was used, and the flow rate was 0.5 mL/min

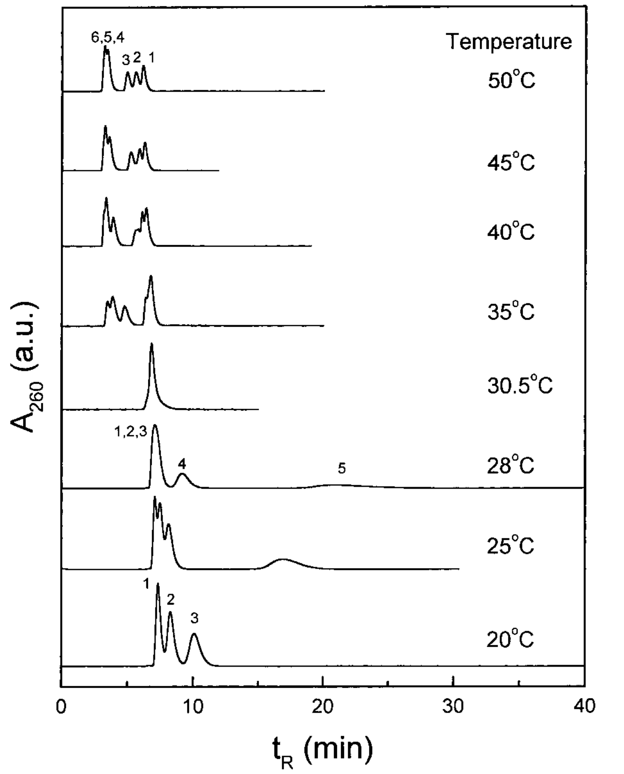

Fig. 3. Temperature effect on the retention of the same 6 PS standards in Fig. 2. The eluent was a mixture of CH_2Cl_2/CH_3CN at the fixed composition of 57/43 (v/v) for all chromatograms. The isothermal chromatograms were obtained at different column temperatures as labeled in the figure. Otherwise, the separation condition is the same as in Fig. 2

weight PS is eluted earlier in the size exclusion separation modes. Since we used a 100 Å pore size column, high molecular weight PS samples (4, 5, 6) are not separated and elute together at the total exclusion limit of the column at the high CH_2Cl_2 composition.

Similar behavior is observed by changing the temperature instead of the eluent composition as predicted by Eq. (2). Fig. 3 shows the temperature effect, where 8 chromatograms taken at different temperatures for the same set of PS standards used in Fig. 2 are displayed. The eluent is a mixture of CH_2Cl_2/CH_3CN at the fixed composition of 57/43 (v/v) for all chromatograms. As we increase the temperature, the retention of PS decreases. This indicates that the sorption process of PS samples to the stationary phase is exothermic as elaborated in the next section. At 28 °C, PS5 (502 k) is eluted near t_R = 20 min as a very broad peak indicating that the isocratic elution of the PS5 is almost impractical in this separation condition. We think that this is in part due to the finite molecular weight distribution of the PS standard in addition to the natural broadening from the mass transfer process. The critical condition is observed at 30.5 °C, consistent with the result shown in Fig. 2. The critical condition is very sensitive to the solvent purity and the composition. As we further increase the temperature, the PS samples are separated mainly by the size exclusion mechanism being eluted before t_m. However, the shape of the SEC chromatograms above the critical temperature gradually changes as the temperature is increased. This is likely due to two effects; (1) the residual contribution of the interaction mechanism and (2) the gradual expansion of the polymer chains as the solvent quality is improved with the increase of the temperature. A similar trend is also shown in Fig. 2, although it is not as clearly visible as in Fig. 3. Small difference in solvent composition near the critical composition, for example, from 55% to 57.5% of CH_2Cl_2 content, changes the elution behavior dramatically as displayed in Fig. 2, while the change is more gradual with the variation of temperature. This is an advantage of the temperature gradient over solvent gradient method, in which fine and more reproducible control is possible. Comparing the retention behavior of the PS standards displayed in Fig. 2 and 3, it is clear that temperature can replace the role of solvent composition to control the retention in the IC separation.

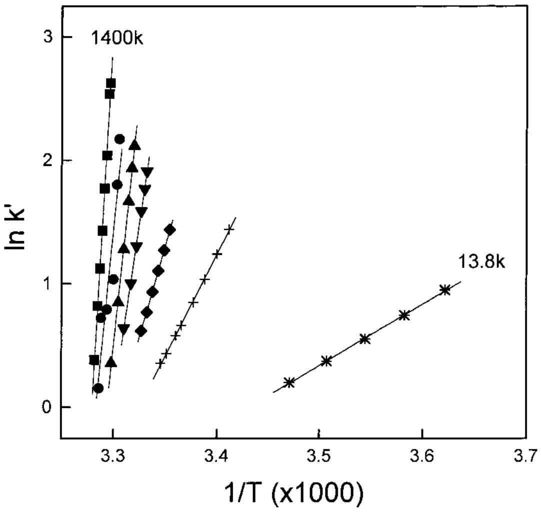

Fig. 4. Temperature dependence of the retention of 7 PS standards ($\bar{M}_w$: 13.8 k, 87.1 k, 208 k, 444 k, 726 k, 1 090 k, and 1 400 k). Separation condition: single C18-bonded phase column (Alltech, Nucelosil, 250 × 2.1 mm, 100 Å pore size, 5 μm particle size); mixed eluent of 57/43 (v/v) CH_2Cl_2/CH_3CN; injection volume of 20 μL; sample concentration of 1 mg/mL each

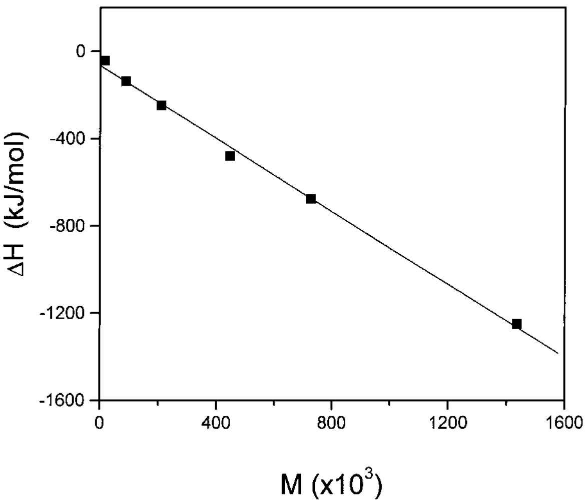

Fig. 5. Molecular weight dependence of ΔH^0 obtained from the slope of the plots in Fig. 4. The slope is equivalent to −87 J/mol-monomer unit

Thermodynamics of the separation mechanism

In the previous section, we showed that the sorption of PS to the stationary phase is exothermic. This is further elaborated from the van't Hoff plots displayed in Fig. 4. In Fig. 4, the $\ln k'$ of 7 PS samples are plotted against $1/T$ where the slope of the plot is ΔH^0 of the sorption process. In this experiment, one C18 bonded silica column with 100 Å pore size and the mixed eluent of CH_2Cl_2/CH_3CN = 57/43 (v/v) were used to measure k' of 7 PS standards at different temperatures. Their molecular weights range from $13.8 \cdot 10^3$ to $1400 \cdot 10^3$. The $1/T$ dependency of $\ln k'$ shows a good linear relationship, and the slope becomes steeper as the molecular weight of PS increases. ΔH^0 is obtained from the slope and plotted as a function of the molecular weight in Fig. 5. ΔH^0 shows a good linear relationship with the molecular weight following the Martin's rule. The slope is equivalent to −87 J/mol-monomer unit. This is a fairly weak exothermic interaction, but it sums up to a large value for a polymer with a high degree of polymerization enough to cast suspicion about the feasibility of isocratic elution of high polymers as mentioned in the introduction section. From this result, we confirm that the IC separation utilizes the sorption process of the polymeric solute onto the stationary phase and the ΔH^0 associated with the sorption process is the control parameter in the IC separation process.

It is worth while mentioning briefly other chromatographic polymer fractionation methods using temperature and/or solvent composition variation here[26,27]. Precipitation chromatography, also called Baker-Williams method, is a method whereby a small quantity of polymers deposited on an inactive support, e. g., glass beads, packed in a column, and successive elution is carried out with a liquid of increasing solvent power. A temperature gradient is set from high to low temperature along the column. In this method, packing material acts as a supporting material and the molecular weight dependent solubility is the main parameter for the fractionation. Temperature rising elution fractionation, often used to fractionate the semi-crystalline polymers in terms of their crystallinity, shares the same concept. Precipitation chromatography can also be carried out in the absence of a temperature gradient – it is then often called as fractional solution method. Solvent gradient HPLC fractionation utilizing the principle of precipitation-redissolution process has been applied successfully for the characterization of copolymers[5].

Solute retention in the chromatographic separation of polymers can occur by either sorption of molecules to the stationary phase or by precipitation-redissolution process[5,28]. Which process applies in a given situation depends on the sample solubility, the strength of the interaction of the sample molecules with the stationary phase and the amount of the sample injected[28]. The characteristic features of two separation mechanisms are well described in the monograph of Glöckner[5]. Precipitation-redissolution mechanism does not seem to play a role for the IC separation of the PS standards described above since all the PS samples are completely soluble at the given separation condition. The cloud point of the highest molecular weight PS is found to be about 0 °C in the mixed eluent.

TGIC and the resolution of TGIC compared with SEC[23]

It is not difficult to imagine how to utilize the temperature effect shown in Fig. 3 for the IC separation of PS.

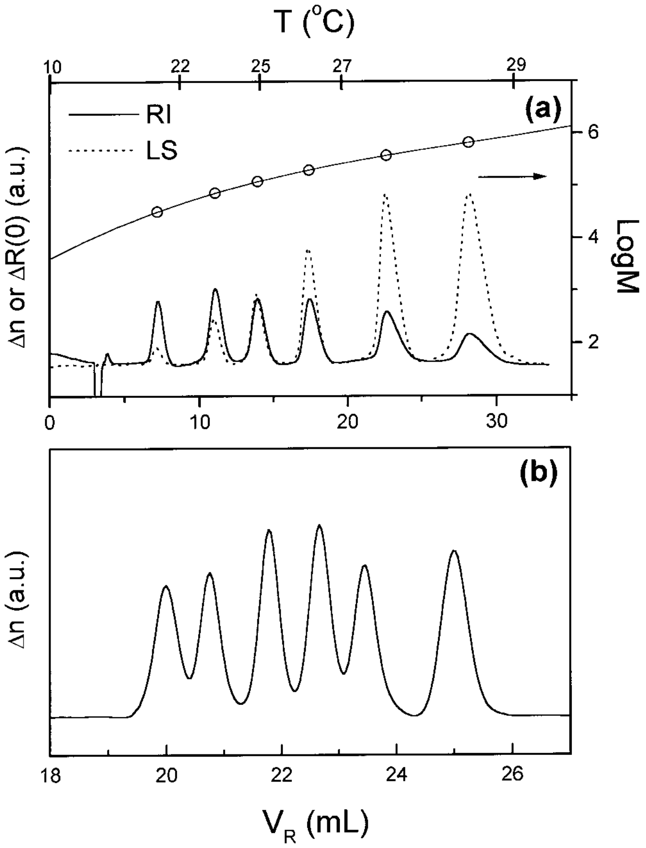

Fig. 6. (a) TGIC chromatograms (solid line: RI detection, dotted line: LALLS detection) of 6 PS standards listed in Tab. 1. The temperature program during the TGIC elution was the 4 segmented linear ramps between 10 °C and 29 °C as shown in the upper abscissa of the plot. The eluent was the same as in Fig. 3, and one C18-bonded silica column (Alltech, Nucleosil, 100 Å pore, 5 µm particle size, 250 ×4.6 mm) was used. The injection sample concentration was 0.5 mg/mL for each PS standard, and the injection volume was 100 µl. The calibration curve (log M vs. V_R) is also shown in the plot. (b) SEC chromatogram of the same PS standards. For the SEC analysis, 4 SEC columns (Polymer Lab, 2 ×Mixed C, 1 ×Mixed D, 1 ×Mixed E) were used and the column temperature was maintained at 40 °C by use of a column oven (FIAtron, CH-460). Eluent was THF, and the flow rate was 0.8 mL/min. Injection sample concentration was 0.5 mg/mL for each polymer in THF and the injection volume was 100 µL

Fig. 6 (a) is a demonstration of the TGIC to separate a mixture of 6 PS standards of which molecular characteristics are listed in Tab. 1. The temperature gradient program during the TGIC elution is shown in the upper abscissa of the plot. The solid line is the chromatogram recorded by a RI detector, while the dotted line is from the LALLS detection. The peak intensity area increases with increasing molar mass for the LALLS detector, as expected for a molar mass sensitive detector. The resolution of TGIC is high enough to fully resolve two standards at baseline whose molecular weights are apart by a factor of 2 or even less. It is hard to achieve a comparable resolution by SEC as demonstrated in Fig. 6 (b) where an SEC chromatogram of the same PS samples is displayed. It is clear from the comparison of two chromatograms in Fig. 6 (a) and (b) that TGIC yields a much higher resolution than SEC at the given column configurations.

Tab. 1. Comparison of the characterization results of PS standards by TGIC and SEC

Code	$\overline{M}_w$ ($\overline{M}_w/\overline{M}_n$)		
	SEC-LS	TGIC-LS	
		(raw data results)[a]	(after 8% correction)
1	32700 (1.02)	35800 (9.5%)	33100 (1.004)
2	74400 (1.02)	79700 (7.1%)	73800 (1.008)
3	127000 (1.02)	133800 (5.4%)	123900 (1.006)
4	213000 (1.02)	226400 (6.3%)	209600 (1.004)
5	394000 (1.02)	425300 (7.9%)	393800 (1.006)
6	683000 (1.03)	753300 (10.3%)	697500 (1.006)

[a] Average deviation: 8%.

For the molecular weight analysis of TGIC, a calibration curve is constructed from the peak molecular weights determined by light scattering detection. A few additional PS standards are used to construct the calibration curve but not shown in the plot. The calibration curve shown in Fig. 6 (a) is obtained from the fitting of the peak molecular weights with a 4th order polynomial. Since we use a mixed solvent, preferential sorption needs to be taken into account for the determination of absolute molecular weight[29]. The details of the correction method will be discussed later. The reproducibility of the retention in the TGIC analysis is sufficient to allow us to use the calibration method relative to the standards of known molecular weight and almost an identical result is obtained[23]. Since it does not provide absolute molecular weights, however, we construct the calibration curve from the LS detection result here. The shape of the calibration curve can be changed easily by altering the temperature program[30]. The temperature program of this TGIC chromatogram was made to exhibit a wide linear range of log M vs. V_R. Using the calibration curve, $\overline{M}_w$ and $\overline{M}_w/\overline{M}_n$ are calculated, and the results are summarized in Tab. 1 together with the results of the SEC analysis. TGIC results of $\overline{M}_w$, corrected for the preferential sorption, are in good agreement with the SEC results for the average molecular weight, while $\overline{M}_w/\overline{M}_n$ values from the TGIC analysis are much smaller than that of SEC analysis.

According to the original treatment of Flory, $\overline{M}_w/\overline{M}_n$ of "*ideal*" anionically polymerized polymer approaches $1 + 1/x$ for high molecular weight polymers, where x is the degree of polymerization[31]. Therefore an anionically polymerized PS with $\overline{M}_w$ of $100 \cdot 10^3$ is supposed to show an ideal $\overline{M}_w/\overline{M}_n$ value of 1.001. To our knowledge this low $\overline{M}_w/\overline{M}_n$ value has never been realized so far. It has

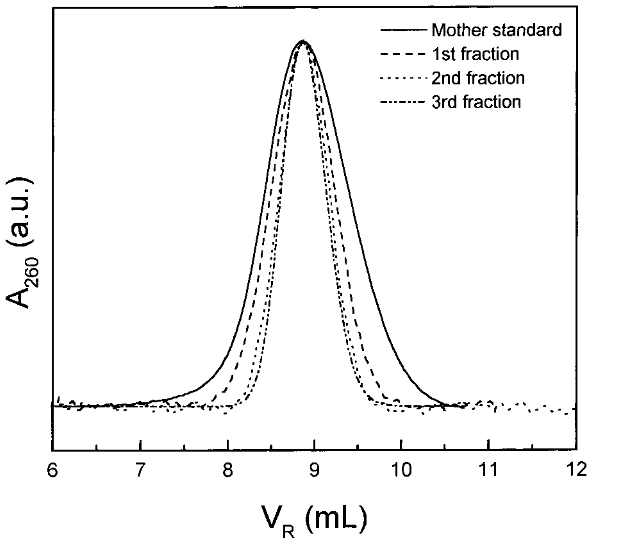

Fig. 7. TGIC fractionation of PS 114 k ($\overline{M}_w = 114 \cdot 10^3$, $\overline{M}_w/\overline{M}_n$ (SEC) = 1.04, $\overline{M}_w/\overline{M}_n$ (TGIC) = 1.004). The elution peak was cut at its half height repeatedly to collect the middle fraction. One column (Alltech, Nucelosil C18, 100 Å pore, 250 × 4.6 mm) was used

been generally believed due to the non-ideal condition of the anionic polymerization and the limited resolution of the analysis technique, in most cases, SEC. Typical $\overline{M}_w/\overline{M}_n$ value of anionically polymerized PS is around 1.01 or higher according to the SEC analysis as shown in Tab. 1. In view of the small $\overline{M}_w/\overline{M}_n$ values obtained from the TGIC analysis in Tab. 1, it is clear that the limitation of SEC resolution is a major cause of the discrepancy. This was previously pointed out from the result of a thermal field flow fractionation experiment[32] and from the analysis of the radius of gyration distribution by SEC coupled with multi-angle laser light scattering detection[33]. SEC is known to suffer from the band broadening, and the band broadening correction is one of the main subjects to be further improved in the SEC data reduction[34].

Since the $\overline{M}_w/\overline{M}_n$ values obtained from TGIC are still significantly larger than the predicted ones from the Flory's prediction, it still remains to be answered whether it is due to the limitation of the TGIC technique itself or due to the non-ideal anionic polymerization condition. In order to find the resolution limit of TGIC, we further fractionate a PS standard. ($\overline{M}_w$ = 114 000, $\overline{M}_w/\overline{M}_n$ (SEC) = 1.04, $\overline{M}_w/\overline{M}_n$ (TGIC) = 1.004) We cut the elution peak at its half height repeatedly to collect the middle fraction of the peak. After fractionating the TGIC elution peak for 3 consecutive times, the peak shape does not change much as shown in Fig. 7. After the fractionation for 4 times, the fractionated sample was subjected to the SEC and TGIC analysis and the chromatograms of the 4th fraction as well as the mother PS are compared in Fig. 8(a) and (b), respectively. TGIC yields a large reduction of $\overline{M}_w/\overline{M}_n$ value from 1.004 for the mother standard to 1.0008 for

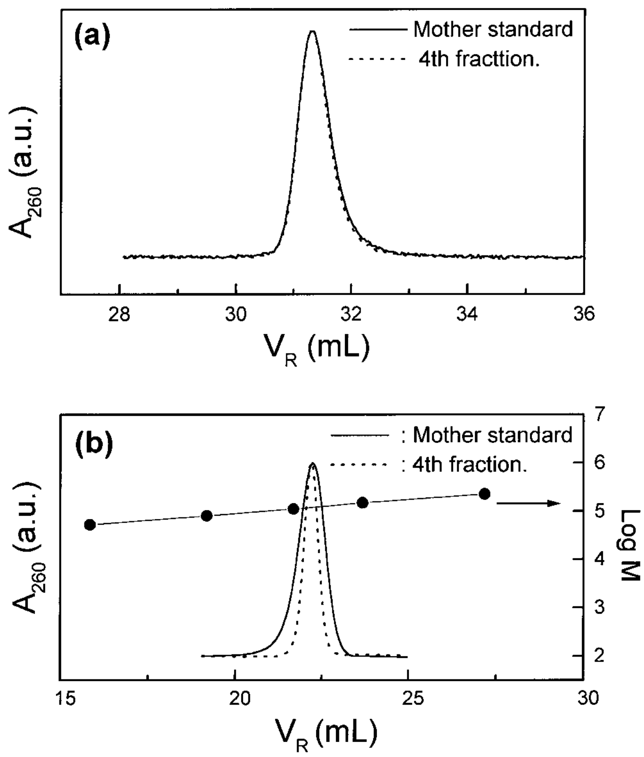

Fig. 8. (a) SEC chromatograms of the mother PS 114 k and its 4th fraction obtained as shown in Fig. 7. (b) TGIC chromatograms of the mother PS 114 k and its 4th fraction. From the calibration curve shown in the plot, the $\overline{M}_w/\overline{M}_n$ of the 4th fraction is determined as 1.0008

the 4th fraction, while SEC does not show a significant change. This result clearly shows that the resolution of TGIC is much higher than the distribution of the mother PS standard, and the $\overline{M}_w/\overline{M}_n$ value of 1.004 for the mother standard obtained by TGIC is believed to be close to the true value. At the same time, it is clear that SEC analysis significantly overestimates the $\overline{M}_w/\overline{M}_n$ value of the PS standards due to the band broadening effect.

TGIC separation of other homopolymers and the effect of mobile phase

In Fig. 9, 3 TGIC chromatograms of PS, polyisoprene (PI) and poly(methyl methacrylate) (PMMA) standards are displayed together with the temperature gradient programs. All three sets of anionically polymerized polymers are well separated by TGIC in terms of their molecular weights. It is important to note that different eluent conditions are employed for each polymer set. As mentioned earlier, temperature alone cannot provide a large enough variation in the interaction strength, so that it is necessary to select a proper solvent system for individual polymer system of interest in order to have temperature play an efficient role to control the retention of the polymers according to their molecular weight. This is a disadvan-

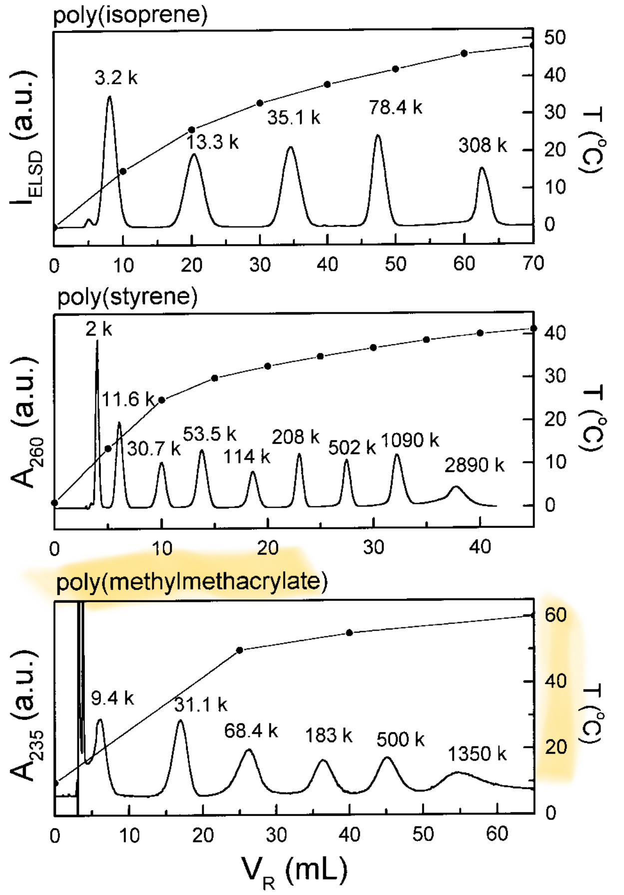

Fig. 9. TGIC chromatograms of polyisoprene (PI), polystyrene (PS), and poly(methyl methacrylalte) (PMMA). Eluent: CH_2Cl_2/CH_3CN mixtures at 80/20 (v/v) for PI, 57/43 for PS, 0/100 for PMMA. Temperature programs are shown in the plots

tage of IC in general compared to SEC, in which a variety of good solvents for a polymer of interest can be used. In order to design an optimum solvent gradient program for the solvent gradient IC separation, solvent strength concept is widely employed[2,35]. It is similar to the concept of solvent quality or solubility parameter, more familiar in the field of polymer science. At the moment, finding a proper solvent system for TGIC relies on empirical tests, however, it is not very difficult for the three polymer systems shown in Fig. 9. All of them are using CH_2Cl_2/CH_3CN mixtures of different compositions. Using a gradient pump, we find the eluent composition at which a polymer of interest is eluted at room temperature. As shown in Fig. 2, it does not span a wide range of solvent composition for a polymer species of different molecular weights, since the retention is very sensitive to the solvent composition. The critical composition of eluent for a polymer is a good guide if it is already reported in the literature. After finding the solvent composition at a given column configuration where the polymer of interest is eluted, fine tuning of the system is done by varying the

Tab. 2. TGIC separation conditions for various polymers

Polymer (mol. wt. range)	Solvent system (volume ratio)	Temperature range
Poly(isoprene) (10^3~10^5)	CH_2Cl_2/CH_3CN (80/20)	5~40 °C
	1,4-Dioxane	20~50 °C
Poly(styrene) (10^2~10^6)	CH_2Cl_2/CH_3CN (57/43)	5~40 °C
	Diethyl malonate	25~40 °C
Poly(methyl methacrylate) (10^3~10^6)	Tetrahydrofuran/H_2O (78/22)	0~40 °C
	CH_3CN	10~60 °C
Poly(vinyl acetate) (10^3~10^6)	CH_3CN/H_2O (91/9)	5~30 °C
Poly(vinyl chloride) (10^3~10^6)	Dimethylformamide	20~50 °C

temperature as well as solvent composition. In Tab. 2 we summarize the TGIC separation conditions for a variety of polymers found in our laboratory.

It is interesting to note that all the eluent systems are near Θ conditions for the corresponding polymers. In order to analyze a wide range of molecular weight in a single TGIC run, interaction with the stationary phase has to be weak, as exemplified earlier (Fig. 5) in the thermodynamic study of PS separation, $\Delta H^0 = -87$ J per mol-monomer segment. Otherwise, high molecular weight polymers would be retained too strongly to be eluted in a reasonable experimental time, since the range of temperature variation is limited due to its freezing(or polymer solubility) and boiling of the solvent system. The stationary phase in the current applications are aliphatic hydrocarbons with limited length solvated with the mobile phase. This stationary phase does not seem to provide a good partition environment for the polymers, since the mobile phase near Θ condition exhibits a small value of ΔH^0 per mol-monomer segment. The detailed mechanism involved in the IC separation process of high polymers needs further investigation, but it is clear that interaction of monomer segment with the stationary phase is a determining factor to optimize the IC separation. It can be controlled by the choice of both mobile and stationary phases. For an example of the mobile phase effect, low molecular weight PS oligomers are known to be well separated with pure CH_3CN eluent, which is a non-solvent for high molecular weight PS[36]. When we measure ΔH^0 of PS oligomers as a function of the degree of polymerization from the van't Hoff plot, we obtain the result shown in Fig. 10. In this case the ΔH^0 per monomer segment is obtained as -2.4 kJ per mol-monomer segment from the slope of the plot, a much stronger interaction than the case of high molecular weight PS shown in Fig. 5. This strong interaction is necessary to separate the oligomers from 6mer to 13mer, but not for the high and wide molecular weight distribution. The stationary phase effect was well demonstrated with PMMA in our previous report that retention of PMMA is much larger with C18-bonded silica than C4-bonded stationary phase[30].

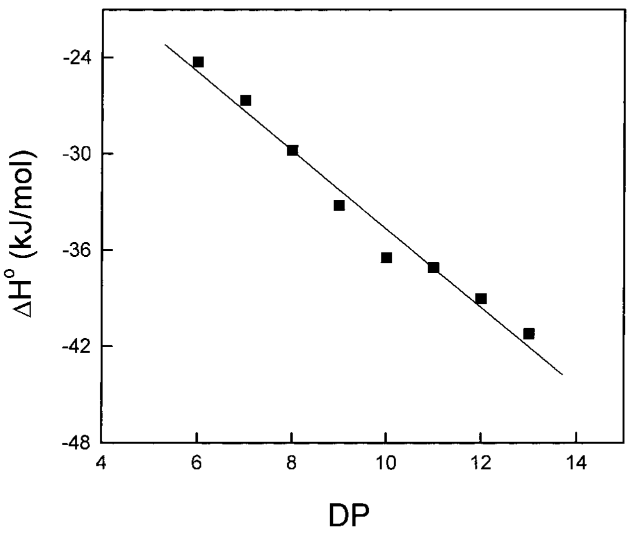

Fig. 10. ΔH^0 vs. degree of polymerization (DP) of PS oligomers from 6mer to 13mer. Stationary and mobile phases were the C18-bonded silica (Alltech, Nucleosil, 100 Å pore) and pure CH_3CN, respectively

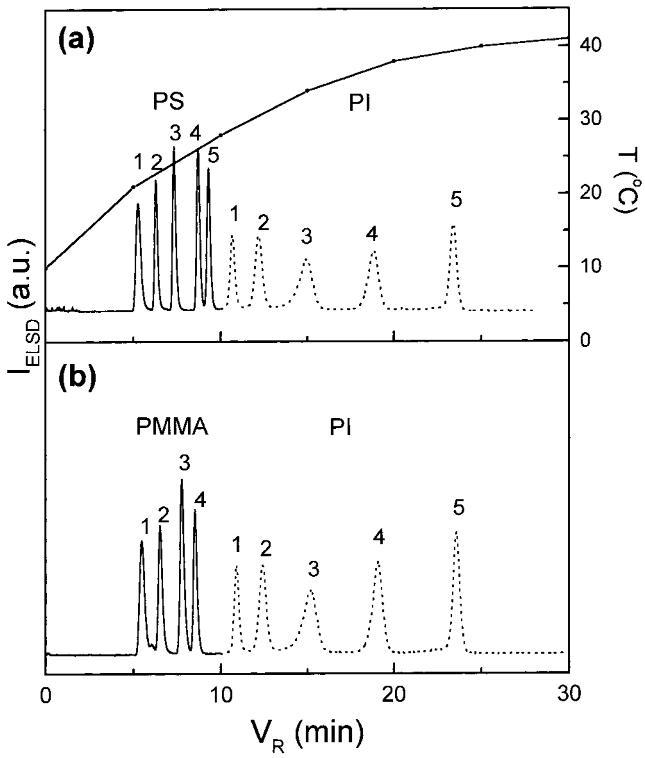

Fig. 11. Simultaneous TGIC and SEC separations of PI/PS (a) and PI/PMMA (b). The identical temperature program as shown in the plot (a) is used for both mixtures. The $\bar{M}_w$ of the polymer standards were, in the sequence of elution; PS: (1) 2890k, (2) 501.5k, (3) 140k, (4) 15.3k, (5) 3.6k, PMMA: (1) 2132k, (2) 364.8k, (3) 78.7k, (4) 17.9k, PI: (1) 2.7k, (2) 10.4k, (3) 24.5k, (4) 53.0k, (5) 200k. The polymer mixtures were made at the concentration of 1 mg/mL for each polymer standard, and the injection volume was 50 μL. We used ELS detector to detect both polymer species

Simultaneous TGIC and SEC of polymer mixtures[37]

In recent years, practical use of polymer mixtures has increased rapidly. Consequently a rapid and precise characterization of their constituents becomes more and more important. Although SEC is the most universal method for polymer characterization, the application of conventional SEC to polymer mixtures is only possible if the polymer chain volumes of the components in a mixture are different enough to be resolved as separate peaks. There have been several approaches to get around this problem. One approach is to use multiple detectors with very different response for the particular component to deconvolute the overlapped peaks[38,39]. Another approach is the coupled LC methods in which the size exclusion mechanism is combined with interaction mechanism. These two different separation mechanisms can be applied sequentially as in orthogonal chromatography[3] or simultaneously as in SEC under critical conditions[25] for the analysis of binary polymer mixtures. Berek et al. developed a more practical method based on the full adsorption of particular components and the desorption by eluent switching to separate polymer mixtures[40,41].

TGIC can be used elegantly employing an isocratic mobile phase and one type of column for the simultaneous characterization of binary polymer mixtures in a single elution. The principle is not complicated. As shown in Fig. 9, PI, PS, and PMMA are well separated by TGIC at different eluent conditions. For an example, consider the eluent composition in the TGIC separation of PS, 57/43 (v/v) mixture of CH_2Cl_2/CH_3CN. It is a good solvent condition for PMMA relative to their TGIC separation conditions. If the eluent composition is good enough for PMMA, it can be separated by the size exclusion mechanism, while PS is being separated by the interaction mechanism, when a mixture of PS and PMMA is injected under the TGIC eluent condition of PS. During the TGIC run, the retention of PMMA does not change much with the variation of the temperature since the size exclusion mechanism is mainly an entropic effect[37].

Fig. 11 is a demonstration of this idea where the simultaneous TGIC/SEC chromatograms of two mixtures of 5 PI and 5 PS standards (a), and 5 PI and 4 PMMA standards (b) are displayed. For this experiment, three C18-bonded silica columns having different pore sizes (Alltech, Nucleosil, 100, 500, 1000 Å pore size each, 5 μm particle size, 250 × 4.6 mm) were connected in series to enhance the resolution of the size exclusion region. The mobile phase was a mixture of CH_2Cl_2/CH_3CN at the composition of 80/20 (v/v), which is the TGIC separation condition of PI in Fig. 9. Temperature profile during the elution is also shown in Fig. 11(a) and the same temperature program is effective in (b). The mobile phase is good enough for PS and PMMA to be separated mainly by the size exclusion mechanism. Therefore, they are eluted before the solvent peak and the higher molecular weight species are eluted earlier than the lower molecular weight ones. The retention volume of the injection solvent peak is about 10 mL, which does not appear clearly due to the ELS detection in these chromatograms.

As displayed in Fig. 11, a perfect separation of the polymer mixtures is achieved by single isocratic elution. As shown in Fig. 9, single C18-bonded silica column is good enough to provide a comparable resolution for PI in the IC region. We employed 3 columns in this study in order to enhance the resolution in the SEC region. Therefore, it is well demonstrated that the mixtures of PI/PS and PI/PMMA can be separated with a good resolution in terms of their chemical nature and molecular weight simultaneously. Our approach shares the same concept as the coupled LC method[25,41] in the sense of using both size exclusion and interaction mechanisms, while our method differs from other coupled LC methods by using both mechanisms simultaneously to separate both polymer components in a single elution.

TGIC with light scattering detection[29,30]

One of the distinct advantages of TGIC over solvent gradient method is that one can have more freedom in choosing detectors. In the analysis of synthetic polymers which often do not contain a chromophore to absorb light, differential refractometer is a more universal detector. With the solvent gradient elution, differential refractometry is difficult to use, if not impossible, due to the background signal drift. Light scattering detection, another useful method in the analysis of synthetic polymers, also relies on the refractive index contrast in the system and is difficult to use with large background drift associated with solvent gradient. While refractive index is also a function of temperature and there must be some refractive index drift in the TGIC elution, temperature can be reduced back, in principle, to a desired constant temperature by the time when the eluent reaches the refractive index sensitive detectors. This is shown earlier in Fig. 6, which demonstrates that temperature is stabilized when the eluent reached the detectors not to show a measurable drift in the TGIC chromatograms recorded by RI detector as well as LALLS detector.

When we use a single component eluent, for example, PMMA analysis with pure CH_3CN eluent, there are no further complications associated with light scattering detection other than controlling the temperature at the detector[30]. When we use a mixed solvent, however, preferential sorption need to be taken into account to analyze the light scattering signal. Polymer chains in a binary mixture solvent preferentially adsorb one component of the mixture, usually the better solvent of two components, so that the refractive index contrast is not only due to the polymer chain itself but also from the excess solvent component preferentially adsorbed to the polymer chain[42]. Therefore an analysis of the light scattering experiment in such a system without considering the preferential sorption effect does not provide a true molecular weight of the polymers.

As shown in Fig. 6(a), the baselines of both RI and LALLS chromatograms are very stable, so that the temperature variation during the elution causes little refractive index drift when eluent reaches the detectors. Complete separation of 6 PS standards is achieved, and the relative intensities of the signal from RI and LALLS detector reflect the molecular weight difference of eluted samples. The raw $\overline{M}_w$ values determined by TGIC/LALLS are summarized together with the SEC/LALLS results in Tab. 1. They show a systematic deviation from the $\overline{M}_w$ values obtained from the SEC/LALLS analysis, which are believed to be the correct values. The percent error shown in the parenthesis, which stands for the deviation from the value of SEC/LALLS measurement, somewhat fluctuates but is approximately constant, $8 \pm 2\%$, independent of molecular weight within experimental precision. This systematic deviation is experimentally confirmed to be caused by preferential sorption[29]. Therefore the correct molecular weights are obtained by multiplying the same correction factor for all PS samples. The corrected molecular weights are in excellent agreement with the values obtained by SEC/LALLS analysis. Undoubtedly, the use of the light scattering detection greatly enhances the applicability of TGIC in the characterization of variety of polymers.

TGIC analysis of branched PS[29]

As mentioned earlier in the introduction section, SEC does not resolve branched polymers as well, since SEC does not separate the polymer molecules in terms of their molecular weight but according to the size of the polymer chain[3]. Branching generally reduces the size of a polymer chain relative to the linear polymer of the equivalent molecular weight[43]. For example, the hydrodynamic volume of star polymers with a uniform arm molecular weight does not change much as the number of arms increases. Therefore it is hard to resolve multi-arm star polymers by SEC as the number of arms increases. Unlike SEC, TGIC utilizes the interaction between polymer segments and the stationary phase; thus TGIC is able to separate branched polymers with greater sensitivity to molecular weight than SEC.

In Fig. 12, the SEC (a) and TGIC (b) chromatograms of 10 star PS samples are displayed. The star PS samples were prepared by linking living PS anions with 1,2-bis(-trichlorosilyl)ethane[44]. A stoichiometry, PS anion/Si-Cl,

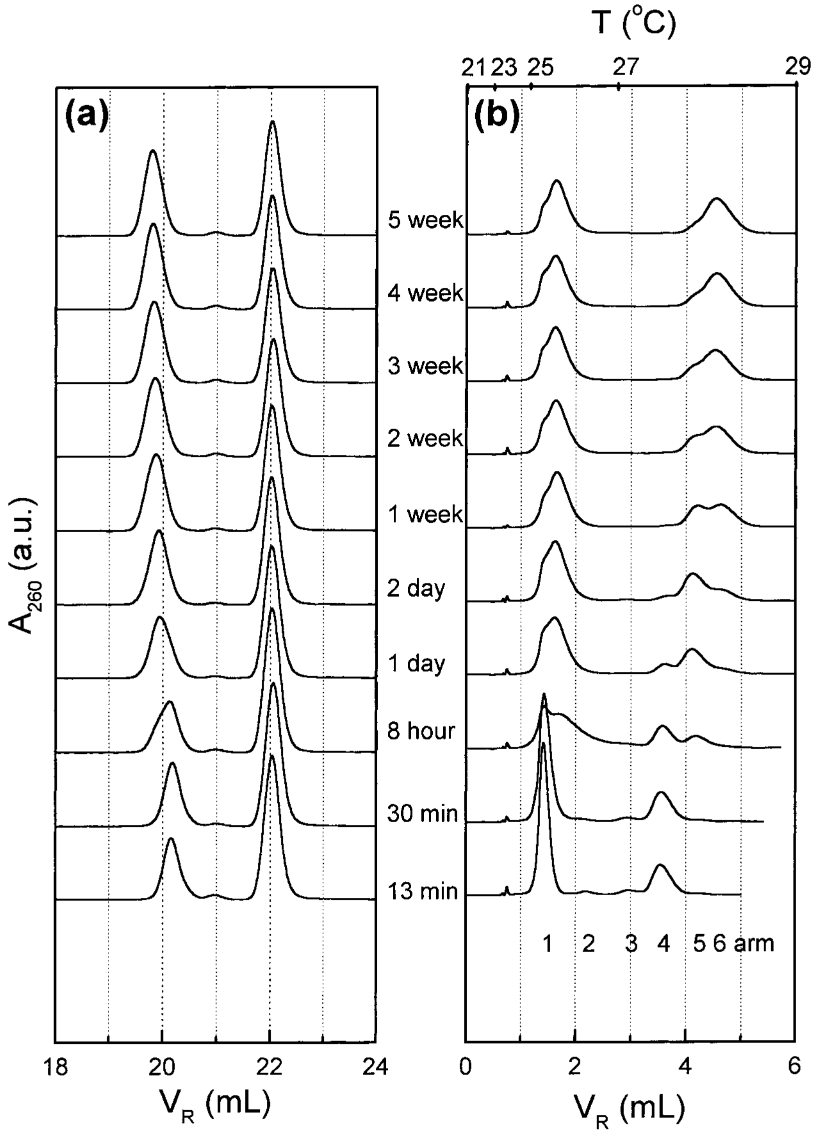

Fig. 12. SEC (a) and TGIC (b) chromatograms of PS star molecules terminated at various time during the linking reaction of living PS anions and 1,2-bis(trichlorosilyl)ethane. The numbers between the SEC and TGIC chromatograms are the linking reaction times allowed prior to termination. For the SEC analysis, star PS solutions were made in THF at a concentration of 0.5 mg/mL and 100 μL of the solution was injected. Four SEC columns (Polymer Lab, 2 Mixed C, 1 Mixed D, 1 Mixed E) were used, and the flow rate was 1.0 mL/min. In the TGIC analysis mobile phase was a mixture of CH_2Cl_2/CH_3CN premixed at the volume ratio of 57/43 and one C18-bonded silica column (Alltech, Nucelosil, 100 Å pore, 250 ×2.1 mm, 5 μm particle size) was used. Sample solutions were made at a concentration of 2 mg/mL, and 20 μL of the solution was injected. The flow rate was 0.1 mL/min. Temperature program of TGIC anaysis is shown in the upper abscissa

of 2/1 was maintained in this reaction. Prior to the linking reaction about five units of isoprene were incorporated onto each living PS chain to reduce steric hindrance during the linking reaction. Small aliquots were taken from the reactor at various linking reaction times and the excess PS anions (unlinked arms) were terminated with degassed methanol. The numbers between the SEC and TGIC chromatograms are the linking reaction times allowed prior to termination.

The SEC chromatogram in Fig. 12 (a) shows that star PS and excess unlinked arms are well separated down to the baseline. As listed in the second column of Tab. 3, $\langle M_{CAL} \rangle$ and $\langle M_{LS} \rangle$ of the unlinked arms are in good agreement with each other and remain constant, independent of the linking reaction time. The angular bracket stands for the weight average quantity of the elution peak, and the subscripts 'CAL' and 'LS' denote the molecular weight determination methods, the calibration against

Tab. 3. Molecular weights (in 10^3 Dalton) of star shaped polystyrenes

Reaction time	SEC		TGIC			
	$\langle M_{LS}\rangle/\langle M_{CAL}\rangle$		$\langle M_{LS}\rangle$	M_{LS}/M_{CAL}		
	arm	star	arm	4	5	6
13 min	84.4/85.0	322/293	81.1	325/324		
30 min	83.1/84.3	322/294	81.1	322/326		
8 h	83.3/85.7	358/317	85.0	313/329	397/396	
24 h	83.0/85.7	393/343	83.1	313/333	394/388	
48 h	82.8/86.3	420/354	83.4		403/389	468/441
1 week	83.8/85.7	440/367	84.7		399/401	472/449
2 week	83.3/84.6	454/368	83.6			487/438
3 week	83.7/84.9	456/373	85.8			483/436
4 week	85.2/85.6	467/378	85.7			487/440

standard linear PS and the light scattering detection, respectively. The small peak appearing near V_R = 21 mL corresponds to star PS with 2 arms (PS_2) whose $\langle M_{CAL}\rangle$ is $170 \cdot 10^3$ Dalton, close to twice the $\langle M_{CAL}\rangle$ of unlinked arms. This small amount of PS_2 exists for all the samples, which may indicate minor coupling reaction between living arms during the work up process. Since the amount is nearly negligible and not involved in the linking reaction process, it is not considered in the analysis of the linking reaction kinetics to be discussed later. Star PS species of three or more arms (PS_3, PS_4, PS_5 and PS_6) are unresolved and eluted together as a single peak near V_R of 20 mL, although V_R of the peak gradually decreases as a longer reaction time is allowed, which indicates the progressive increase of the average molecular size. As listed in the third column of Tab. 3, both $\langle M_{LS}\rangle$ and $\langle M_{CAL}\rangle$ of the peak increase with increased linking reaction time. However, $\langle M_{LS}\rangle$ and $\langle M_{CAL}\rangle$ are no longer identical, and the deviation between the two becomes larger and larger as the reaction time increases. As mentioned earlier, this is an expected behavior of the SEC analysis, in which the polymer chains are separated in terms of their hydrodynamic volume. Since the hydrodynamic volume of the star polymer is smaller than the linear polymer of the equivalent molecular weight, M_{CAL} is smaller than M_{LS}. Light scattering analysis is supposed to provide a true weight-average molecular weight, and $\langle M_{LS}\rangle$ of $322 \cdot 10^3$ at 13 min reaction time indicates that PS_4 forms very fast in the early stage of linking reaction. $\langle M_{LS}\rangle$ finally reaches the value close to the molecular weight of the 6 arm star, $469 \cdot 10^3$, after the reaction for several weeks, while $\langle M_{CAL}\rangle$ reaches an asymptotic value at $380 \cdot 10^3$.

In Fig. 12(b) the TGIC chromatograms of the same set of star PS samples are displayed. Temperature is programmed to increase in 4 segments of linear ramp between 21 °C to 29 °C as shown in the upper abscissa of the plot. Since we are dealing with a relatively narrow molecular weight range, the small temperature change is

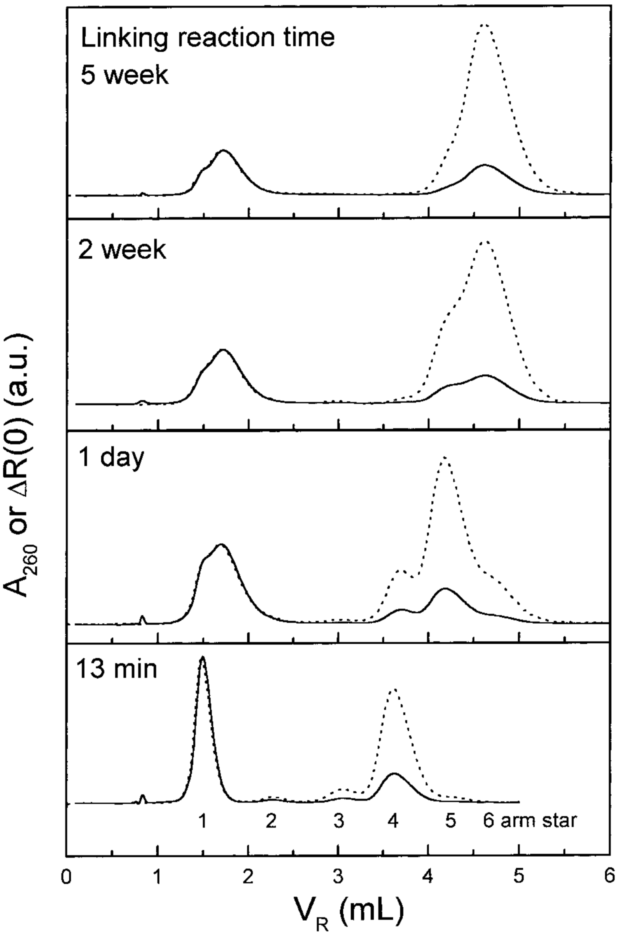

Fig. 13. TGIC chromatograms of PS stars terminated at 4 linking reaction times recorded by UV/Vis (solid line) and LALLS (dotted line) detector

sufficient in this analysis. The injection solvent peak appears at $V_R \cong 0.8$ mL. It is evident that TGIC provides far better resolution than SEC. The chromatogram of the polymer mixture obtained after 13 min linking reaction time clearly shows 4 resolved peaks at $V_R \cong 1.5$ mL (unlinked arm), 2.2 mL (2 arm), 3.0 mL (3 arm) and 3.6 mL (4 arm). 5 and 6 arm star molecules are yet to be developed clearly at this short reaction time. It is clearer in the bottom plot of Fig. 13 (13 min) where two chromatograms recorded by UV/Vis (solid line) and LALLS (dotted line) detectors are displayed. The star species in each elution peak can be easily identified from the relative intensities from two detection methods. Although it is not clear in the UV/Vis recording, in the LALLS recording the existence of a small amount of 5 arm stars can be found as a tail of the 4 arm star peak due to its high molecular weight. In Fig. 12(b), 5 arm stars become clearly visible in the UV/Vis chromatogram at $V_R \cong 4.2$ mL in the chromatogram of 8 h linking reaction time, while the development of 6 arm stars becomes clear only after the reaction for a few days at $V_R \cong 4.7$ mL.

Interestingly enough, the elution peak of the unlinked arms starts to split to form a new broad peak at larger V_R. The peak splitting and broadening is clearly displayed at V_R around 1.7 mL in most of the chromatograms except for the two at the early stage of linking reaction. This peculiar behavior is not observed in the SEC analysis. As shown in the equivalent SEC chromatograms in Fig. 12(a), the V_R and the peak width of the unlinked arms remain practically unchanged during the whole reaction period. The identity of the new peak can be easily inferred from the comparison of the UV/Vis and LALLS chromatograms in Fig. 13. The traces of the split peak recorded by two different detection methods are well matched, which indicates the molecular weights of the polymers in the split peak are identical. When we calculated $\langle M_{LS} \rangle$ of the whole split peak, as shown in Tab. 3, the $\langle M_{LS} \rangle$ does not change significantly with the peak broadening and is practically the same as the $\langle M_{LS} \rangle$ of the unlinked arms from the SEC analysis. A small difference between the normal ($81 \cdot 10^3$) and broadened peak ($85 \cdot 10^3$) appears to come from the contribution of 2 arm species, since the broadened peak overlaps with the PS_2 peak. In addition, the relative amount of the unlinked arms in SEC (unimodal peak at $V_R = 22$ mL) and in TGIC (the broad bimodal peak at 1.2 mL $< V_R <$ 2.5 mL) are identical as elaborated later (Fig. 15). Therefore the broadened peak in TGIC represent the unlinked arms. It seems that the single arm species have different end group to which TGIC is sensitive while SEC is not. The nature of these chain ends is not known but likely to be developed during the procedure of taking the aliquot from the reaction mixture[45].

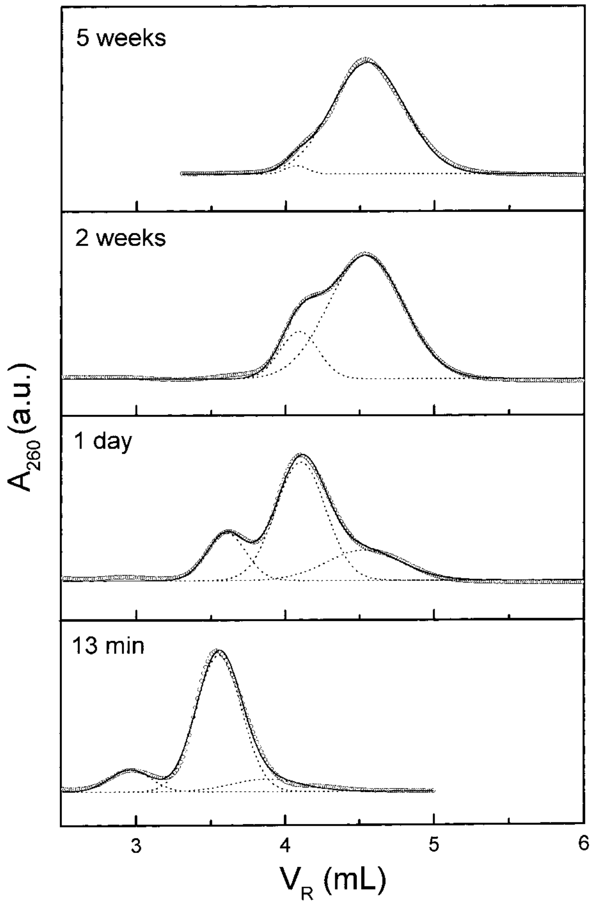

Fig. 14. Separation of PS_3, PS_4, PS_5 and PS_6 from the expanded plot of the UV/Vis chromatogram in Fig. 13. Dotted lines are the isolated Gaussian peaks of PS_4, PS_5, or PS_6, and the solid line is the sum of the isolated peaks, which is in good agreement with the chromatogram itself (open circles)

Linking reaction kinetics of PS star[45]

Although the resolution of TGIC is much better than SEC, it is still not possible to separate the elution peaks of PS_4, PS_5 and PS_6 completely. Since the peaks are not fully resolved, it is difficult to obtain a precise average molecular weight of each species. Instead, we determined M_{CAL} and M_{LS} at the peak positionsof PS_4, PS_5, and PS_6. The results are summarized in the last 3 columns of Tab. 3. Unlike the SEC analysis, M_{CAL} and M_{LS} for PS_4, PS_5, or PS_6 are much closer each other and provide the molecular weights expected from the number of arms in each species. In order to obtain the relative abundance of each star species, we tried to isolate each peak as shown in Fig. 14, where expanded chromatograms of PS_3, PS_4, PS_5 and PS_6 are displayed. Fitting to multiple Gaussian peaks was carried out, and the result is displayed in Fig. 14, where the dotted lines show the isolated Gaussian peaks of PS_4, PS_5, or PS_6 and the solid line is the sum of the isolated peaks, which is in good agreement with the chromatogram itself (open circles). From the areas of separated peaks, the fractional abundance of PS_1, PS_2, PS_3, PS_4, PS_5 and PS_6 normalized by the initial concentration of living arms at each linking reaction time can be obtained by Eq. (3).

$$P_i = \frac{A_i / \langle M_{LS} \rangle_i}{\sum_{i=1}^{6} A_i / \langle M_{LS} \rangle_{arm}} \quad \text{and} \quad \sum_{i=1}^{i=6} i \cdot P_i = 1 \tag{3}$$

where P_i is the relative number abundance of PS_i, and A_i and $\langle M_{LS} \rangle_i$ are the elution peak area and $\langle M_{LS} \rangle$ of PS_i, respectively. Since peak area from the UV detection is proportional to mass concentration, it has to be divided by the molecular weight of the corresponding species to obtain a molar concentration. Thus obtained time-varying relative abundance of each species are displayed in Fig. 15. In Fig. 15, relative amounts of the unlinked arm, P_1 is obtained from both TGIC and SEC chromatograms and plotted with filled (SEC) and open (TGIC) circles. They are in good agreement with each other, confirming

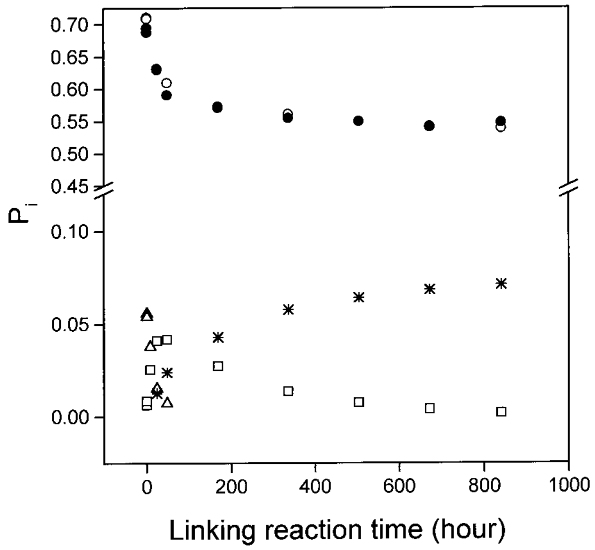

Fig. 15. Change of relative abundance of PS_4 (triangle), PS_5 (square), PS_6 (star) and unlinked arm (open circle) during the linking reaction. The relative population of unlinked arm determined by SEC analysis is also plotted as filled circles

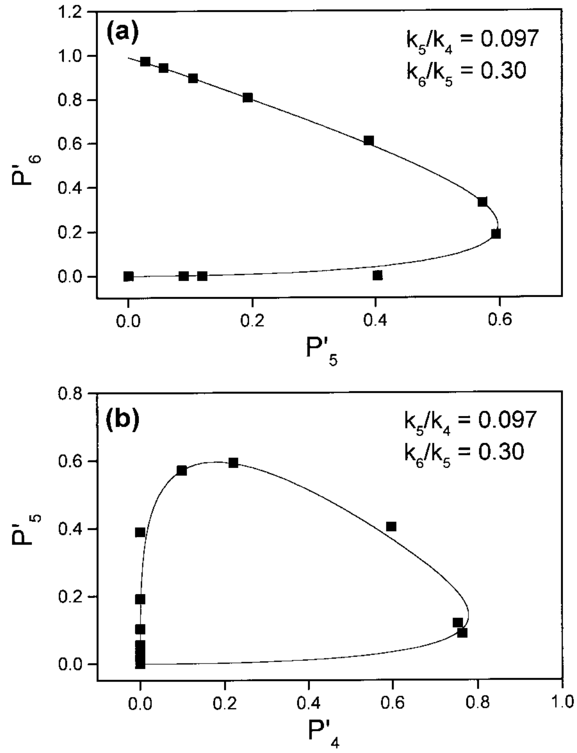

Fig. 16. Change of relative population of PS_6' vs. PS_5' (top), and PS_5' and PS_4' (bottom) during the linking reaction. Solid squares represent the relative amount of respective star species normalized by the total amounts of the star species. Solid lines are the best fit to the data points according to the consecutive linking reaction model with the floating parameters of k_5/k_4 and k_6/k_5

that the partially split broad peak in TGIC indeed represents the unlinked arm. The concentration of unlinked arm decreases rapidly at the early stage of the linking reaction, but the reaction rate slows down considerably at the later part of the reaction. This is an expected behavior considering the difficulty of incorporation of the 5th and 6th arms due to steric hindrance.

From the time-varying relative abundance of each species with different number of arms, we can extract quantitative information on the linking reaction rate for the late stage[45]. Since the initial linking reaction to form PS_2 and PS_3 is much faster than the "synthetic time constant", i.e., time required to take an aliquot from the reactor and to terminate the linking reaction, it is impossible to follow the early stage of the linking reaction. The result is shown in Fig. 16(a) and (b), where the plots of P_6' vs. P_5' and P_5' vs. P_4' are shown, respectively. P_i' denotes the P_i normalized by the total amount of the star species, i.e., the total amount of the chlorosilane compounds. Considering the experimental difficulty associated with the high vacuum technique and the deconvolution to individual Gaussian peaks, the quality of the fit appears quite reasonable.

Thus obtained ratios of the rate constants are k_5/k_4 = 0.097 and k_6/k_5 = 0.30. This result indicates that the linking reaction becomes slower as the number of attached arms increases; the reaction rate of incorporating the 5th arm to PS_4 is 10 times slower than linking the 4th arm to PS_3, while the reaction rate of incorporating the 6th arm to PS_5 is 3 times slower than attaching the 5th arm to PS_4. The decreasing trend of the reaction rate with the increase of the number of attached arms is as expected; however, it is interesting to note that k_6/k_5 is significantly larger than k_5/k_4. This trend is clearly shown in Fig. 16. At their maximum abundance, PS_4 reaches about 80% of total star population, while PS_5 only reaches about 60%. Although it appears at a glance in contradiction to the expected trend, we believe that the molecular structure of the linking agent can provide a clue to explain this observation. The 6 chlorosilane functional groups do not all reside within a single sphere of steric interaction; rather, they exist as two identical trichlorosilane groups connected by a short spacer, each representing a separate branching point. Considering steric hindrance near the branching points, PS_4 likely has a structure of 2 + 2 arm star, i.e., 2 arms are attached to each trichlorosilane group. As far as the steric hindrance in the vicinity of the chlorosilane group is concerned, k_5 and k_6 represent the identical reaction, i.e., linking the anionic end of a living arm polymer to the last chlorosilane group left in a trichlorosilane group, while k_4 is the rate constant of incorporating a living arm to one of the two remaining chlorosilane groups. The experimental result appears to reflect this difference of the linking reaction well.

Sample loading capacity of TGIC

In order to study the various physical properties of synthetic polymers, it is often required to acquire the samples with narrow molecular weight distribution. Polymers with wide molecular weight distribution are usually fractionated either by fractional precipitation or fractional solution methods[26,27]. These methods can supply moderate quantity of the fractionated samples, however, the resolution is not high enough to provide narrow fractions of which $\overline{M}_w/\overline{M}_n$ is comparable to anionically polymerized polymers. The only way to obtain fractions with such a narrow distribution is chromatographic separation. However, size exclusion chromatography is not an efficient tool to fractionate a polymer sample in a preparative scale due to its small sample loading capacity. We investigated the sample loading capacity of TGIC to evaluate the potential of TGIC to be further developed as a preparative fractionation method.

In Fig. 17, the sample loading capacity of SEC and TGIC is compared by injecting various amounts of polymer samples. In order to examine the change of the resolution with the sample loading, mixtures of 9 PS standards at different total concentrations are injected and their chromatograms are compared. The column configuration is quite different in two methods; 4 columns (Alltech, Jordi DVB gel, 250 mm × 10 mm) were used in SEC, while 1 column (Alltech, Nucleosil C18, 250 mm × 4.6 mm, 100 Å) was used in TGIC. Therefore, in terms of the physical volume of the columns, SEC columns have a total interior volume about 18 times as large as that of TGIC. The figures in the plots represent the amount of each PS standard injected. As the amount of injection is increased, the retention in TGIC (Fig. 17 (a)) gradually decreases and the peak shape deteriorates from the ideal Gaussian shape. The decrease of the retention volume is mainly due to the saturation of the stationary phase. It is another indication that the retention in the TGIC fractionation occurs not by the precipitation-redissolution but by the sorption of the polymer chains to the stationary phase, since the retention generally increases with the sample size in the precipitation-redissolution mechanism[5]. Although the resolution becomes poor at high sample loading, the PS standards are reasonably well separated up to 2 mg loading of each PS standard (18 mg of total loading). The injection volume was 50 μL, and the concentration of the injection sample was about 30% at the highest loading. Considering the viscosity of the injection sample and the small size of the column, the resolution is remarkable. This high loading capacity of TGIC becomes more conspicuous when we compare it with the SEC separation. As displayed in Fig. 17(b) the resolution of SEC deteriorates rapidly with the increase of the sample loading. When 0.5 mg of each PS standards are injected, the resolution of high molecular weight samples is completely lost. Therefore IC appears to have strong potential to be further developed as a preparative method to obtain narrowly dispersed polymer samples.

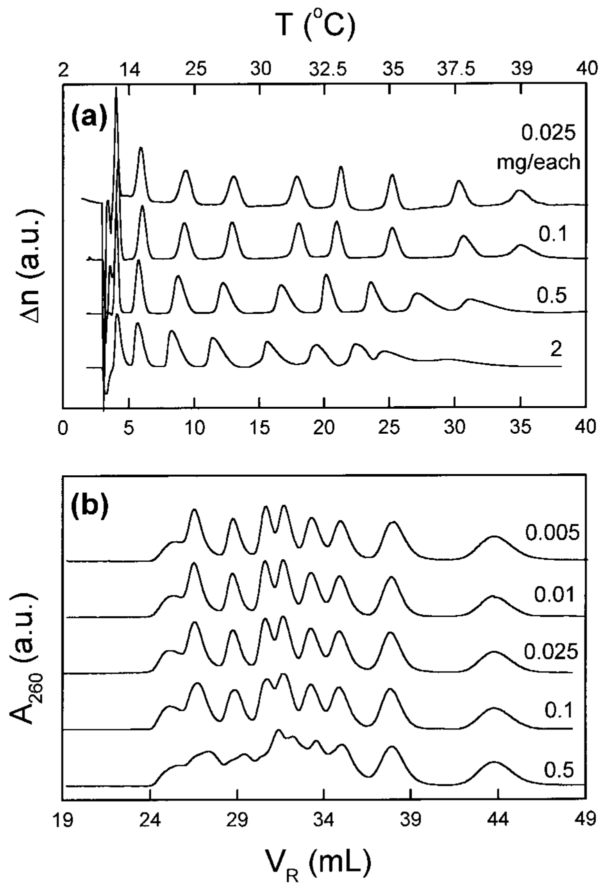

Fig. 17. The comparison of the sample loading capacity in TGIC (a) and SEC (b). Temperature program for TGIC is shown in the upper abscissa. The numbers on each chromatograms stand for the amount of each PS standard injected. Separation conditions are: For TGIC, 1 column (Alltech, Nucleosil C18, 250 mm × 4.6 mm, 100 Å), mixed eluent of CH_2Cl_2/CH_3CN (57/43, v/v), 50 μL of injection volume. For SEC, 4 columns (Alltech, Jordi DVB gel, 250 mm × 10 mm), THF eluent, 50 μL of injection volume

Remarks on the future development

We close our report by pointing out the merits and shortcomings of this new method. As demonstrated earlier, by virtue of its high resolution, IC is useful to rigorously characterize the molecular weight distribution of narrowly dispersed polymers. SEC is not able to provide the correct molecular weight distribution of the polymer standards made by anionic polymerization due to its band broadening effect. In addition, the sample loading capacity of IC is much higher than SEC. Therefore it appears promising for a small scale preparative purpose of narrowly dispersed polymers. Also its ability to fractionate the star shaped polymers with high resolution shows a good potential for the analysis of branched polymers. It could be made more effective if it is coupled with SEC since two techniques separate the polymer molecules by different mechanisms. In addition, the isocratic elution in TGIC allows us more freedom in choosing detectors such as refractive index detector or light scattering detector which are not so useful in the solvent gradient IC method.

On the other hand, IC is not as universal as SEC in a sense that an optimized condition has to be found for each polymer to be characterized. It includes the right choice of stationary phase, mobile phase, temperature gradient, and so on. If one needs a precise evaluation of the molecular weight distribution or works with polymers whose optimum TGIC conditions are known, this new method provides an edge over the conventional SEC. We do not expect TGIC is going to compete with SEC in the polymer molecular weight characterization in general. SEC is certainly superior in the facility and universality to IC. We rather expect TGIC would fill the weakness of SEC for some specific applications such as the analysis of branched polymers.

Despite its rather long history of IC on the molecular weight characterization of polymers, application to the practical problems in the polymer molecular analysis is barely started. In order to exploit the full potential of IC in the molecular characterization of polymers, further studies on the detailed separation mechanism are necessary, which could lead to the development of more efficient methods. Most of IC separations so far reported in the literature have been carried out in near Θsolvent condition with aliphatic hydrocarbon bonded silica stationary phase, which sometimes causes difficulty due to the polymer solubility. In addition, the pore size effect is not fully understood in the IC separation of polymers. Unlike the separation of small molecules, size exclusion mechanism inevitably plays an important role in the partition process due to the large size of polymer molecules if the porous stationary phase is employed. We expect a more predictable application of IC for the polymer characterization in the near future.

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